0400CST428052303

Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

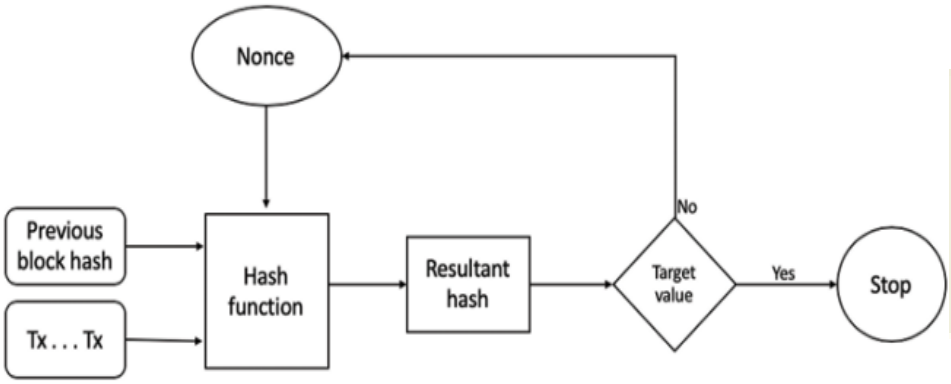
Eighth Semester B.Tech Degree (R,S) Examination May 2024 (2019 Scheme)

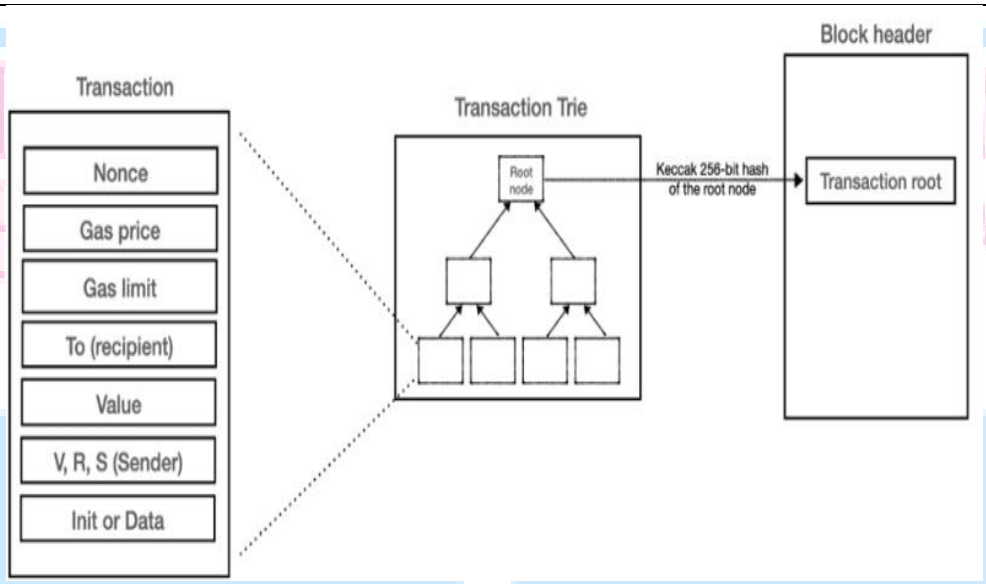
Course Code: CST428

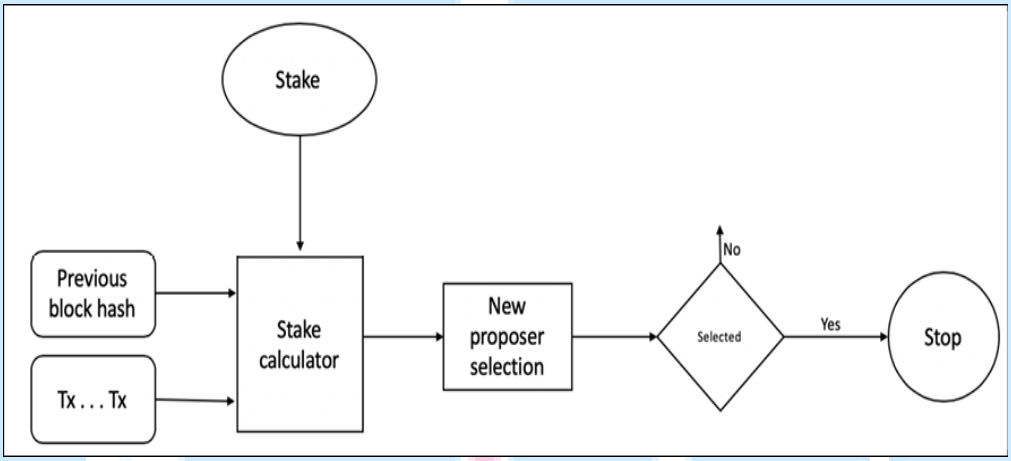
Course Name: BLOCK CHAIN TECHNOLOGIES

Max. Marks: 100		Duration: 3 Hours

PART A

		Answer all questions, each carries 3 marks.	Marks
1		Any three differences with examples (each 1 mark)	(3)
2		Any three benefits (each 1e mark)	(3)
3		1. Proof-based, leader-election lottery-based, or the Nakamoto consensus 2. Byzantine fault tolerance (BFT)-based	(3)
4		Features like - Ownership, Architecture , Security, High availability, Fault tolerance, Collusion resistance, Trust etc (their comparison)	(3)
5		1. That transaction inputs are previously unspent. 2. That the sum of the transaction outputs is not more than the total sum of the transaction inputs. 3. That the digital signatures are valid, which ensures that the script is valid.	(3)
6		 Working of PoW	(3)
7		Definition – (1 marks) Any four properties (Each 0.5 mark)	(3)
8		Explanation -3 marks	(3)

9			(3)
10		By using the CALLCODE or DELEGATECALL opcode of the EVM.	(3)
		PART B	
		Answer onefull question from each module, each carries 14 marks.	
		Module I	
11	a)	Idea and purpose – (2 marks) RSA algorithm – (5 marks) Example – (2 marks)	(9)
	b)	Properties – 2 marks Working of DHT with an example – 3 marks	(5)
		OR	
12	a)	Logic of Ellipitical curve – 2 mark ECC key exchange – 3 marks ECC encryption -2 marks ECC decryption -2 marks Advantages – 1 marks	(10)
	b)	Diagram of SHA-256 compression – 4 marks	(4)
		Module II	
13	a)	Working of block chain (4 marks) Diagram (3 marks)	(7)
	b)	idea of disintermediation (3.5 marks) idea of competition (3.5 marks)	(7)
		OR	

14	a)	Storage(2 marks), communication (2 marks), computation (3 marks)	(7)
	b)	Benefits (3 marks), features (2 marks) and limitations (2 marks) of block chain	(7)
		Module III	
15	a)	Working of Paxos (5 marks) Benefits (3 marks)	(8)
	b)	PoS working idea (4 marks) Figure (2 marks)  	(6)
		OR	
16	a)	Steps for bitcoin transcations 7 marks	(7)
	b)	Definition – (2 marks) Types – (5 marks)	(7)
		Module IV	
17	a)	Idea of Oracle – (3 marks) Generic data flow – (4 marks)	(7)
	b)	Applications like border control, voting , citizen identification (7 marks)	(7)
		OR	
18	a)	Explanation – 7 marks	(7)
	b)	DApp -3 marks. Design -4 marks	(7)
		Module V	
19	a)	Definition – (2 marks) Message call (2.5 marks) Contract creation (2.5 marks)	(9)

	b)	Explanation- 5 marks	(5)
		OR	
20	a)	Any 3 control structures each (2 marks)	(6)
	b)	Contract on bank process (8 marks)	(8)
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